

processing conducted securely in the cloud. The startup is working towards SOC2 certification standards and states that users retain control over the data they choose to share. The wearable links to a user's email ID and uses voice fingerprinting to ensure that only authorised individuals can access stored data. Even if the device is lost, the data remains protected through account-level security.

While the startup works with specialised vendors for component manufacturing, final quality control takes place in Bengaluru. "Most of the team's effort, however, is concentrated on software, artificial intelligence, and backend infrastructure rather than hardware assembly," explains Dhananjay.

The startup currently operates with an 11-member team. Production capacity allows the manufacturing of up to 5000 devices per month. The wearable is priced at ₹12,000, includes a two-year warranty,



(R) Dhananjay Yadav and (L) Aryan Yadav, Founders, NeoSapien Private Limited

and comes with a companion mobile application for setup and integration.

The startup primarily targets entrepreneurs, business owners, busy professionals, journalists, and on-field sales teams.

Revenue is currently driven by device sales and expanding B2B deployments. The company is also

exploring licensing its proprietary Neo OS to larger hardware manufacturers, indicating a potential shift from device manufacturing to AI operating system development.

Dhananjay states that the name NeoSapien reflects a new phase in human evolution. Although the creation of information has grown rapidly in recent years, human

memory has not kept pace. The product is positioned as a tool to augment human intelligence rather than replace it.

Future plans include expanding desktop applications and adding advanced automation features such as calendar integration and pre-read generation before meetings.

—Nitisha Dubey

2 REIMAGINING PRE-SILICON QUALITY AND SECURITY FOR INDIA'S RISC-V ERA

As the RISC-V ecosystem expands, the startup advances a clear premise: hardware innovation must be matched by equal progress in trust, quality, and security.

Reimagining pre-silicon quality and security for India's RISC-V era, Vyoma Systems is a Chennai-based deep-tech startup founded in 2021 and incubated at IIT Madras Pravartak. Headquartered in Chennai and operating as a private limited firm, the company focuses on one of the semiconductor industry's most complex and underestimated challenges: pre-silicon verification and security.

At the core of its offering is UpTickPro, a unified, cloud-enabled pre-silicon platform that integrates functional verification, FPGA-based validation, early software bring-up, and side-channel security analysis. By adopting a

Python-based methodology over traditional SystemVerilog UVM flows and costly EDA toolchains, the platform reduces regression cycles, accelerates coverage closure, embeds security earlier in the design lifecycle, and reportedly halves verification time. Agentic AI workflows further enhance regression analysis and coverage review, enabling small teams to deliver outcomes comparable to larger verification groups.

The company was founded to address a critical gap in the growing RISC-V ecosystem. As startups and enterprises increasingly adopt domain-specific and custom processor architectures, verification remains costly, time-intensive, and

reliant on large engineering teams. For emerging Indian design houses, this creates barriers in cost, time-to-market, and technical complexity. The founders identified that the primary bottleneck in chip innovation lies not only in architecture, but in achieving functional correctness, faster coverage closure, and robust pre-silicon security.

The name 'Vyoma,' derived from Sanskrit for 'sky' or 'space,' reflects the company's ambition to build a secure and expansive foundation for India's semiconductor ecosystem. Its core innovation lies in shifting verification and security earlier in the design lifecycle. A key differentiator is its pre-silicon side-channel